(FP1) 1. Roots of polynomial equations

NO. 1

The cubic equation $x^3 + bx^2 + cx + d = 0$, where $d \neq 0$, has roots α, β, γ such that $\gamma = \frac{1}{\beta}$.

a. Show that $\beta + \frac{1}{\beta} = d - b$. [3]

b. Show also that $\beta + \frac{1}{\beta} = \frac{1-c}{d}$. [2]

c. It is given that $b = 3$, $c = -3$ and $d > 0$.

i. Find the value of d. [2]

ii. Find the value of $\alpha^2 + \beta^2 + \gamma^2$. [2]

NO. 2

The quartic equation $x^4 + x^3 + x^2 + x + 1 = 0$ has roots $\alpha, \beta, \gamma, \delta$.

a. Show that a quartic equation with roots $2\alpha + 1, 2\beta + 1, 2\gamma + 1, 2\delta + 1$ is $y^4 - 2y^3 + 4y^2 + 2y + 11 = 0$. [4]

The sum $(2\alpha + 1)^n + (2\beta + 1)^n + (2\gamma + 1)^n + (2\delta + 1)^n$ is denoted by S_n.

b. Find the value of S_2. [2]

c. Given that $S_3 = -22$, find the value of S_4. [2]

NO. 3

The cubic equation $x^3 + 2x + 1 = 0$ has roots α, β, γ.

a. Find a cubic equation whose roots are $\alpha^3 - 1, \beta^3 - 1, \gamma^3 - 1$. [3]

b. Find the value of $(\alpha^3 - 1)^2, (\beta^3 - 1)^2, (\gamma^3 - 1)^2$. [2]

c. Find the value of $(\alpha^3 - 1)^3, (\beta^3 - 1)^3, (\gamma^3 - 1)^3$. [2]

NO. 4

It is given that

$$\alpha + \beta + \gamma + \delta = 2,$$
$$\alpha^2 + \beta^2 + \gamma^2 + \delta^2 = 3,$$
$$\alpha^3 + \beta^3 + \gamma^3 + \delta^3 = 4,$$

a. Find the value of $\alpha\beta + \alpha\gamma + \alpha\delta + \beta\gamma + \beta\delta + \gamma\delta$. [2]

b. Find the value of $\alpha^2\beta + \alpha^2\gamma + \alpha^2\delta + \beta^2\alpha + \beta^2\gamma + \beta^2\delta + \gamma^2\alpha + \gamma^2\beta + \gamma^2\delta + \delta^2\alpha + \delta^2\beta + \delta^2\gamma$. [3]

c. It is given that α, β, γ, δ are the roots of the equation

$$6x^4 - 12x^3 + 3x^2 + 2x + 6 = 0.$$

i. Find the value of $\alpha^4 + \beta^4 + \gamma^4 + \delta^4$. [3]

ii. Find the value of $\alpha^5 + \beta^5 + \gamma^5 + \delta^5$. [2]

NO. 5

The cubic equation $2x^3 + x^2 - px - 5 = 0$, where p is a positive constant, has roots α, β, γ.

a. State, in terms of p, the value of $\alpha\beta + \beta\gamma + \gamma\alpha$. [1]

b. Find the value of $\alpha^2\beta\gamma + \alpha\beta^2\gamma + \alpha\beta\gamma^2$. [2]

c. Deduce a cubic equation whose roots are $\alpha\beta$, $\beta\gamma$, $\alpha\gamma$. [1]

d. Given that $\alpha^2 + \beta^2 + \gamma^2 = \frac{1}{3}$, find the value of p. [2]